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THE FIELD EFFECT TRANSISTOR

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1. INTRODUCTION TO THE FET TRANSISTOR

The **Field Effect Transistor**, or **FET**, is a three terminal unipolar semiconductor device which uses a conductive central channel to control the flow of current through itself by the application of different voltages onto its three terminals. These terminals are labelled the: **Source** (**S**), the **Drain** (**D**), and the **Gate**, (**G**) respectively.

There are two main types of field effect transistor, the *Junction FET* (**JFET**), and the *Insulated-gate FET* known commonly as a **MOSFET** (Metal-Oxide Semiconductor Field Effect Transistor).

Junction FET's are very similar in their construction to MOSFET's except they have a semiconductor junction area under the gate region known as a depletion layer. Both types of field effect transistor use the voltage applied to their input terminal, the Gate, to control the current flowing through them resulting in the output current being proportional to the input voltage.

As the electrical resistance of the main conductive channel relies heavily on the electric field (hence the name field effect) generated by the input Gate voltage, this then makes the Field Effect Transistor a "VOLTAGE" controlled device (unlike the BJT which is a current controlled device).

The Field Effect Transistor consists of a single conducting channel of doped silicon which behaves like a voltage controlled resistance. One end of this conductive channel is called the **Source** terminal, since current is sourced into it. While the other end of the conductive channel is called the **Drain** terminal, as electric current is drained away from it.

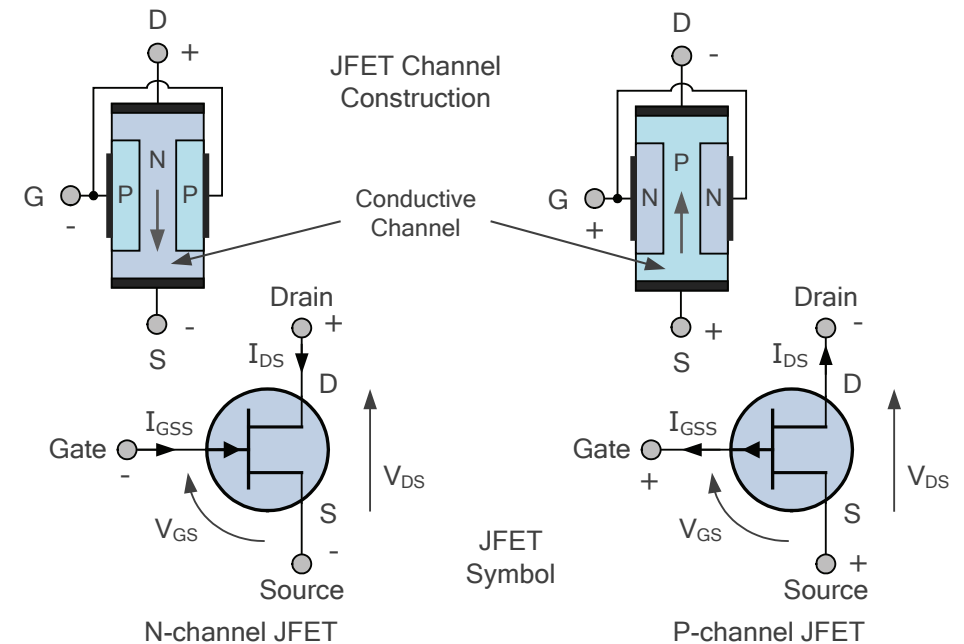
Field Effect Transistors are voltage controlled devices which have a very high input impedance

The electrical current which flows from Drain to Source will depend on the resistive value of the channel, as well as the Drain to Source voltage V_{DS} . Thus, if the FET's channel is doped evenly along its length, then its conductivity will be the same at all points along it. Then the $I \cdot R$ voltage drop created along the channel will be linear with respect to the source terminal.

There are two basic configurations of field effect transistor known commonly as the **N-channel FET** and the **P-channel FET**. The *N-channel FET* channel is doped with donor impurities meaning that the flow of current through the channel is negative (hence the term N-channel). Likewise the *P-channel FET* is doped with acceptor impurities meaning that the flow of current through the channel is positive (hence the term P-channel).

The construction and circuit symbols used for the N-channel FET and the P-channel FET are shown in Figure 1.

FIGURE 1. THE JUNCTION FIELD EFFECT TRANSISTOR



The N-channel JFET has a much greater channel conductivity (lower channel resistance) than their equivalent P-channel types, since electrons have a greater mobility through a conductor compared to holes. This makes the N-channel JFET a more efficient conductor compared to their P-channel counterparts.

Analogies of terminals and voltages between the unipolar Field Effect Transistor and the Bipolar Junction Transistor are given in the following table:

TABLE 1. COMPARISON BETWEEN FET AND BJT TRANSISTORS

Field Effect Transistor	Bipolar Junction Transistor
Source (S) →	Emitter (E)
Drain (D) →	Collector (C)
Gate (G) →	Base (B)
Drain Supply (V_{DD})	Collector Supply (V_{CC})
Drain Current (I_D)	Collector Current (I_C)
Gate Bias Voltage (V_G)	Base Bias Voltage (V_B)

For a Junction Field Effect Transistor (JFET) to operate correctly, it requires two external biasing voltages. One voltage source, V_{DS} is applied between the source and the drain terminals allowing current to flow through the conductive channel.

The other voltage source, V_{GS} is applied between the gate and the source terminals. The gate-source voltage, V_{GS} biases the gate terminal varying the conductivity of the channel and hence the amount of current flowing through it.

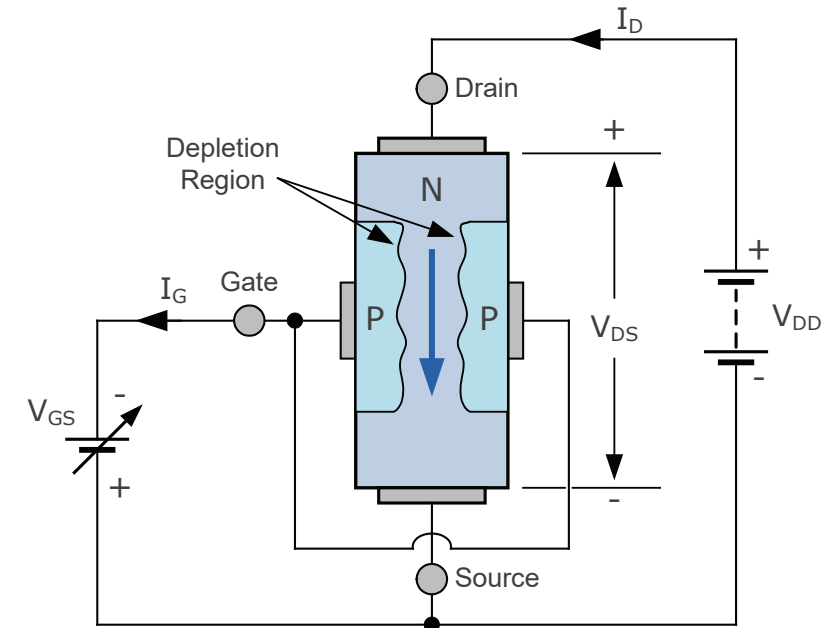
Figure 2 shows that for an N-channel JFET, a P-type region forming the Gate is diffused into the N-type channel to create a reverse biased pn-junction.

It is this semiconductor pn-junction which forms the depletion region around the Gate area which should not be forward biased. That is when no external voltages are applied.

Note also that the depletion region of Figure 2. is much wider at the drain end of the channel because the drain-to-source voltage adds to the gate-to-source voltage, creating a higher reverse bias voltage than that appearing across the source end of the channel.

Field Effect Transistors are available in two basic formats, the N-channel FET and the P-channel FET

FIGURE 2. N-CHANNEL JFET BIASING VOLTAGES AND CURRENTS



When the gate terminal is shorted to the source terminal so that $V_{GS} = 0$, the depletion region becomes very shallow thus the N-channel JFET operates in saturation mode allowing maximum drain current, ($I_D = \text{MAX}$) through the fully open conductive channel.

Thus the N-channel JFET acts as a “closed switch” passing current. The maximum channel current is called the N-channel JFET’s saturation current, I_{DSS} (Drain-to-Source-Saturated-Current). Any JFET data sheet will list this maximum I_{DSS} value.

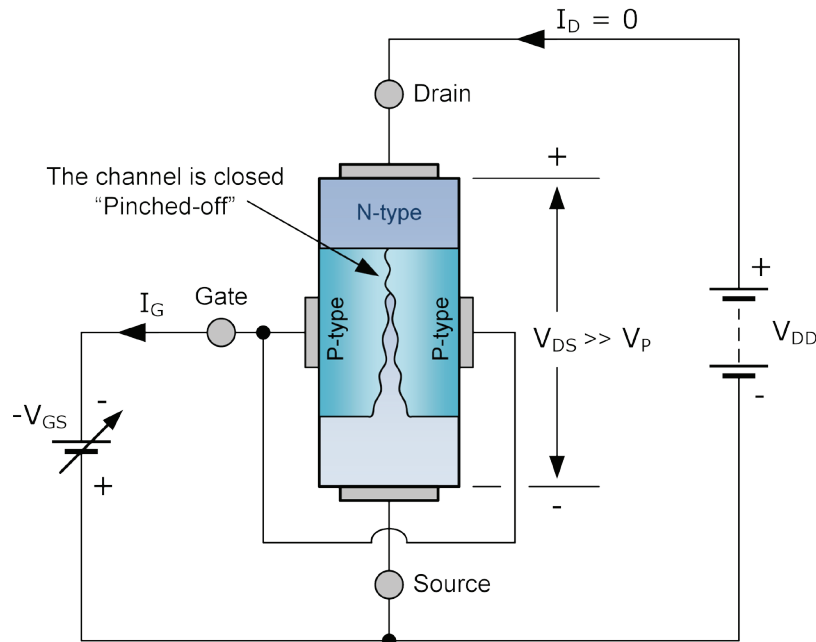
On the other hand, when the gate region is made more negative with respect to the drain, the reverse-bias condition causes the depletion layer to increase in size and extend across the channel changing its width as shown in Figure 3.

The result is a change in the channel resistance between the drain and source limiting current flow through the channel, a sort of “squeezing” effect takes place as the channel is reduced or depleted. Thus JFETs are known as “depletion-type” devices.

If the gate terminal is made progressively more negative the depletion layer expands even more reducing further the effective cross-sectional area of the channel and eventually closing or “pinching-off” the conducting channel entirely (similar to the cut-off region for a BJT). The N-channel JFET now acts as an “open switch” preventing the flow of current.

The voltage point at which this happens is called the threshold or pinch-off voltage, V_p . Thus for an N-channel JFET, the size of the depletion region is controlled by V_{GS} , and as V_{GS} becomes more negative, so too does the depletion region.

FIGURE 3. N-CHANNEL JFET “PINCHED-OFF”



Note that the P-channel JFET is a complementary version of the N-channel JFET so has the same operating characteristics as the N-channel JFET. The main difference being in the direction of the drain current (I_D) through the channel, since in a P-channel JFET, the polarity of the bias voltages, V_{GS} and V_{DS} are opposite to that in an N-channel JFET.

For a typical JFET device, the relationship between drain current, I_D and the gate-source voltage, V_{GS} is given by:

$$I_D \cong I_{DSS} \left[1 - \frac{V_{GS}}{V_P} \right]^2$$

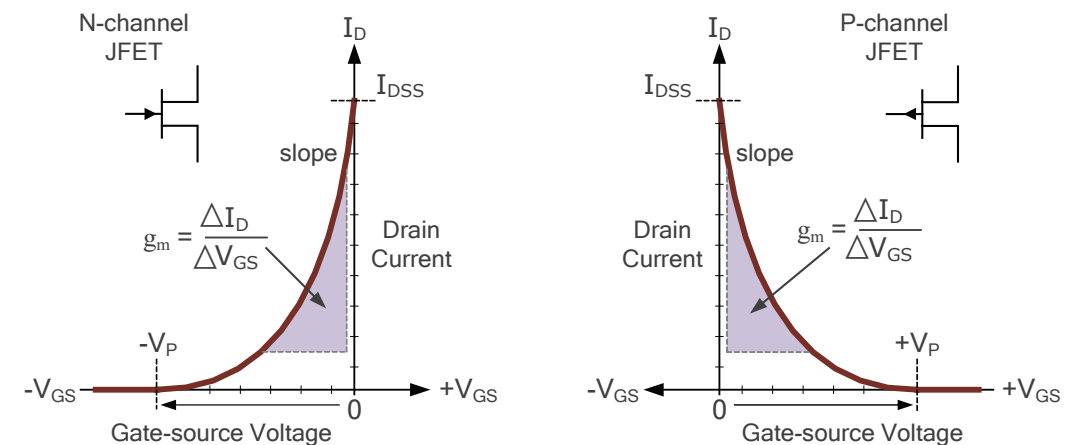
Thus when $V_{GS} = 0$ and V_{DS} is much greater than V_p the channel is saturated (fully-open) and $I_D = I_S = I_{DSS}$. At pinch-off, $V_{GS} = V_p$ so the conductive channel is fully closed and cut-off then no drain current flows and $I_D = 0$.

Remember that V_p will be *Negative* for N-channel JFET's and *Positive* for P-channel JFET's.

2. THE JFET'S TRANSCONDUCTANCE CURVES

As a JFET controls the current flowing through its conductive channel using a voltage applied to its gate, it is therefore a “transconductance” (transfer-conductance) device. The relationship between input voltage to the output current is given the symbol, g_m . The unit of transconductance is the siemens (symbol S), and is equivalent to Amps per Volt, (A/V).

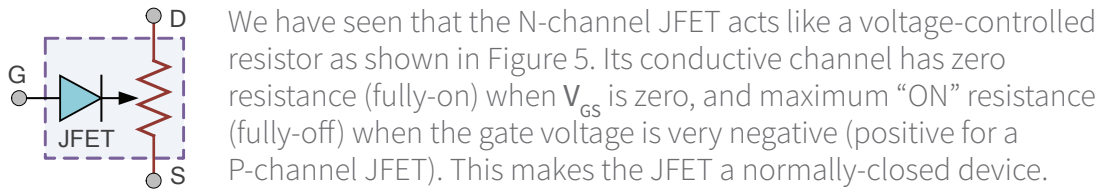
FIGURE 4. TRANSCONDUCTANCE CURVES



By plotting the value of I_D against V_{GS} for a fixed value of V_{DS} , we can create what is commonly called a transconductance curve of a JFET as shown in Figure 4. for both the N-channel and P-channel JFET types.

Notice that the slope is nonlinear since it follows the previous I_D equations square law. So drain current, I_D increases faster due to the opening of the channel as V_{GS} approaches zero ($V_{GS} = 0$). Then for a JFET, changes in V_{GS} do not produce equal changes in I_D .

FIGURE 5. JFET EQUIVALENCE



By having specific values of gate-source voltage, V_{GS} applied to the gate and varying the voltage between the drain and source, V_{DS} , we can create a set or family of output characteristics curves for a JFET device. These I-V curves are similar to those of the bipolar transistor, showing the relationship between I_D and V_{GS} for different values of V_{DS} .

FIGURE 6. JFET OUTPUT CHARACTERISTICS CURVES

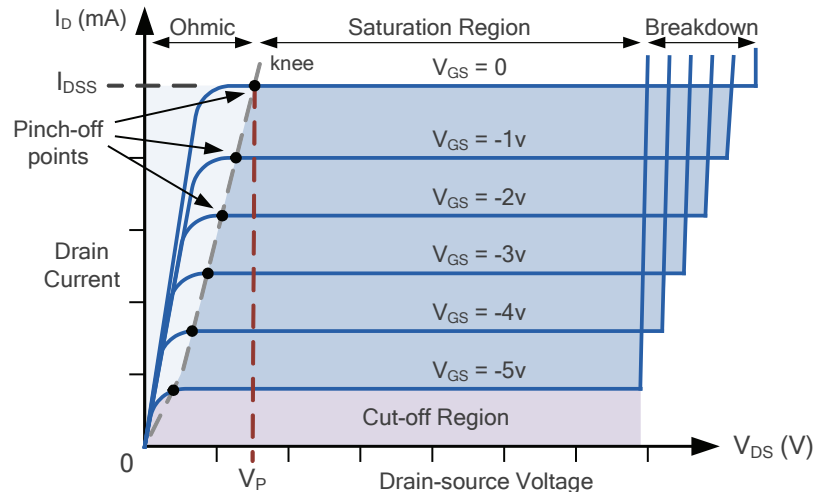


Figure 6. shows the JFET's major operating regions. The shaded area to the left or below pinch-off is called the Ohmic Region and the shaded area to the right or above pinch-off is called the Saturation Region. We can define the four areas or regions of operation of the Junction Field Effect Transistor as being:

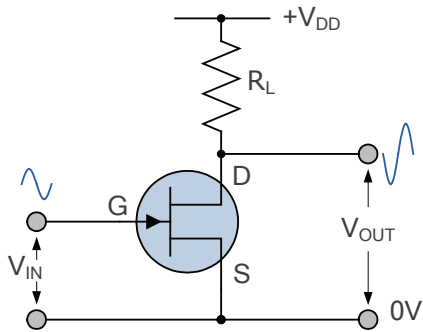
- 1) Ohmic Region** – When $V_{GS} = 0$ the depletion layer of the channel is very small. As V_{DS} increases the current flowing through the channel increases linearly. The JFET acts like a voltage controlled resistor until the knee point is reached at the pinch-off point.
- 2) Cut-off Region** – This is where the gate voltage, V_{GS} is sufficiently negative to cause the channel to close completely and is independent of V_{DS} . The JFET acts as an open circuit with the channel resistance at its maximum so no drain current flows, $I_D = 0$.
- 3) Saturation or Active Region** – The JFET becomes a good conductor with the channel being controlled by V_{GS} resulting in linear amplification. Variations in the drain-source voltage, V_{DS} has little or no effect on changes in channel current, I_D acting as a constant-current source but changes with V_{GS} . When biased in this region, the JFET acts as an amplifier.
- 4) Breakdown Region** – The drain-source voltage V_{DS} is high enough to causes the JFET's resistive channel to break down because the reverse-biased pn-junction is damage. This allows an uncontrolled amount of current to flow through the channel. Thus a very small change in V_{DS} will cause very large uncontrolled change in I_D .

Note then that for fixed values of V_{GS} , increasing the value of the drain-source voltage causes the JFET to operate between Ohmic region (voltage controlled resistor), Saturation region (constant-current source) and Breakdown region (constant-voltage source). For normal amplifier operation, V_{GS} is biased to be somewhere between V_p and 0.

3. THE JFET'S MODES OF OPERATION

Like the bipolar junction transistor (BJT), the field effect transistor (FET) being a three terminal device is capable of three distinct modes of operation and can therefore be connected within a circuit in one of the following configurations.

3.1 Common Source (CS) Configuration

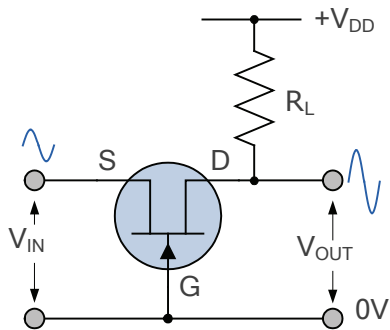


In the **Common Source** configuration (similar to the common emitter amplifier), the input is applied to the gate and its output is taken from the drain.

This is the most common mode of operation of the JFET. This is due to its high input impedance and good voltage amplification. This JFET configuration can be used in many applications where a solid state switching circuit is required in both analogue and digital circuits.

The common source mode of the JFET connection also offers high gain and low noise making them ideal as the basis for audio frequency amplifiers and in high input impedance pre-amps and stages. Common Source amplifiers are widely used but being an amplifying circuit, the output signal is 180° “out-of-phase” with the input signal.

3.2 Common Gate (CG) Configuration



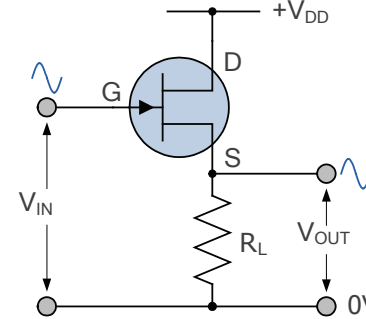
In the **Common Gate** configuration (similar to the common base amplifier), the gate is common between both the drain and source terminals. The input signal is applied to the source and its output is taken from the drain with the gate connected directly to ground (0V).

The high input impedance feature of the previous connection is lost in this common gate configuration as it has low input impedance, but high output impedance. The advantage is that as the gate is

grounded it provides a high level of isolation between the input and output.

This type of FET configuration can be used in high frequency circuits or in impedance matching circuits where a low input impedance needs to be matched to a high output impedance. The output signal is “in-phase” with the input signal.

3.3 Common Drain (CD) Configuration



In the **Common Drain** configuration (similar to the common collector amplifier), the drain terminal is common with the input signal applied to the gate and its output is taken from the source.

The common drain or “source follower” configuration has a high input impedance and a low output impedance with near-unity voltage gain, so are commonly used as buffer amplifiers.

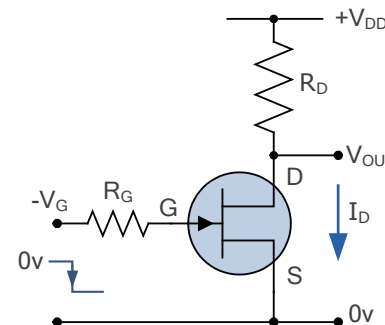
The voltage present, $+V_{DD}$ provides the required bias.

The voltage gain of the source follower configuration is less than unity so the output signal is always “in-phase”, (0°) with the input signal.

4. DC SWITCHING APPLICATIONS

As with the bipolar junction transistor, the JFET being a voltage-controlled device can be used to provide either DC or ON-OFF switching control of a load current. By having a constant supply voltage, V_{DD} and using a gate control voltage, V_G we can switch the channel OFF and ON between its cut-off and saturation regions respectively.

FIGURE 10. JFET SWITCHING CIRCUIT



For a JFET to work as a switch in its common source configuration, the following conditions apply:

$$I_D = \frac{V_{DD} - V_{DS}}{R_D}$$

The JFET will be “fully-ON” when: $V_G = 0$ and $I_D = I_{DSS}$

The JFET will be “fully-OFF” when: $V_G \geq V_p$ and $I_D = 0$

Remember that, V_p is negative for N-channel JFETs, and positive for P-channel JFETs.

5. THE METAL OXIDE FIELD EFFECT TRANSISTOR

As well as the previous *Junction Field Effect Transistor*, there is another type of field effect transistor available which has no gate-to-channel pn-junction available. Instead the gate terminal is electrically insulated from the main current carrying channel by means of a thin insulating silicon dioxide (SiO_2) layer. This type of FET device is therefore called an **Insulated Gate Field Effect Transistor** or **IGFET**.

The most common type of insulated gate FET which is used in many different types of electronic circuits and applications is called the **Metal Oxide Semiconductor Field Effect Transistor** or **MOSFET** for short. Just like the previous JFET device, MOSFET's are available with either N-type or P-type resistive channels.

However, unlike the Junction FET which is available as an N or P-channel depletion type device, MOSFET's are available as both Depletion and Enhancement type devices. Both of these with either N or P-type conductive channels. This makes the MOSFET available in one of four different types of devices.

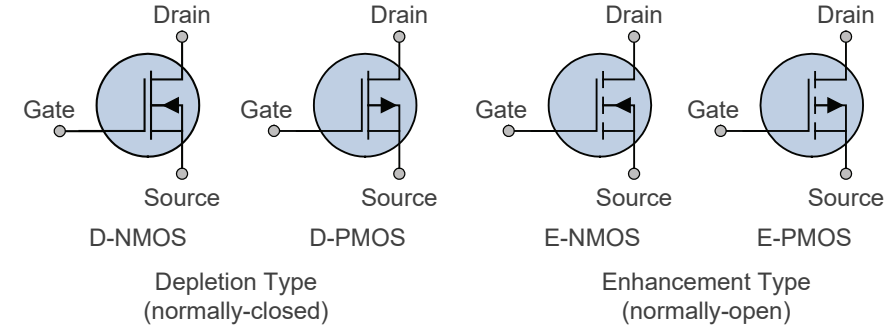
- 1). N-channel Enhancement MOSFET or E-NMOS FET
- 2). P-channel Enhancement MOSFET or E-PMOS FET
- 3). N-channel Depletion MOSFET or D-NMOS FET
- 4). P-channel Depletion MOSFET or D-PMOS FET

MOSFET's have their gate terminal insulated from the conducting channel by a thin layer of insulating metal oxide

The basic difference is that Depletion types are normally-closed devices and a negative gate voltage has to be applied to turn them "OFF". While Enhancement types are normally-open devices so a positive voltage is applied to the gate turn them "ON". Thus the biasing of the gate with respect to the source terminal varies for each MOSFET type.

MOSFETs are three terminal devices with a Gate, Drain and Source the same as for the JFET. Sometimes a 4th terminal, known as the body is also available. The schematic symbols used to identify the four basic variations of the MOSFET's described above are given in Figure 11.

FIGURE 11. MOSFET SCHEMATIC SYMBOLS



The line in the MOSFET symbol between the drain (D) and source (S) connections represents the transistors type of semiconductive channel. If the channel line is a solid unbroken line then it represents a "Depletion" (normally-closed) type of MOSFET as drain current can flow along the solid line with zero gate biasing potential.

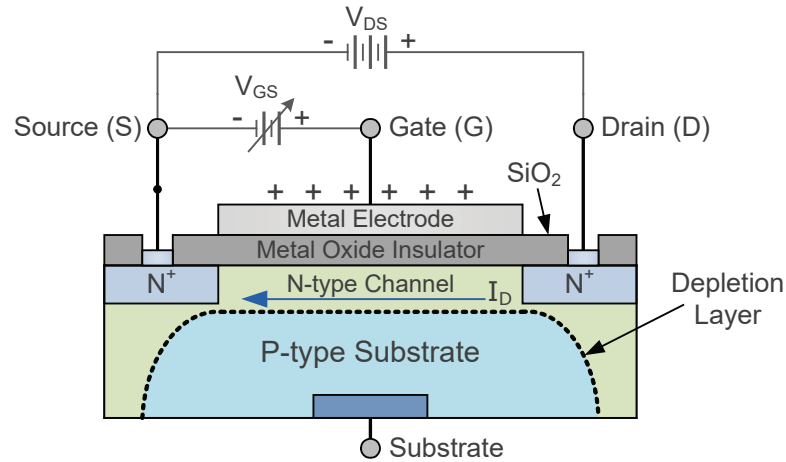
If the channel line is a dotted or broken line, then this represents an "Enhancement" (normally-open) type of MOSFET as zero drain current can flow along a broken line with zero gate potential. The direction of the arrow pointing to this channel line indicates whether the conductive channel is a P-type or an N-type semiconductor device similar to the JFET device.

6. MOSFET CONSTRUCTION

A MOSFET structure is constructed using a lightly doped p-type substrate with two heavily doped n-type materials creating the MOSFET's *Source* (S) and the *Drain* (D) regions. The gap between them being the channel. This device is known as an NMOS or an N-channel MOSFET.

A thin layer of metal-oxide insulating material between source and drain is covered by a conductive material, forming the *Gate* (G) terminal. Thus giving its physical construction of being a Metal on top of an Oxide on top of a Semiconductor, hence the name MOS as shown in Figure 12.

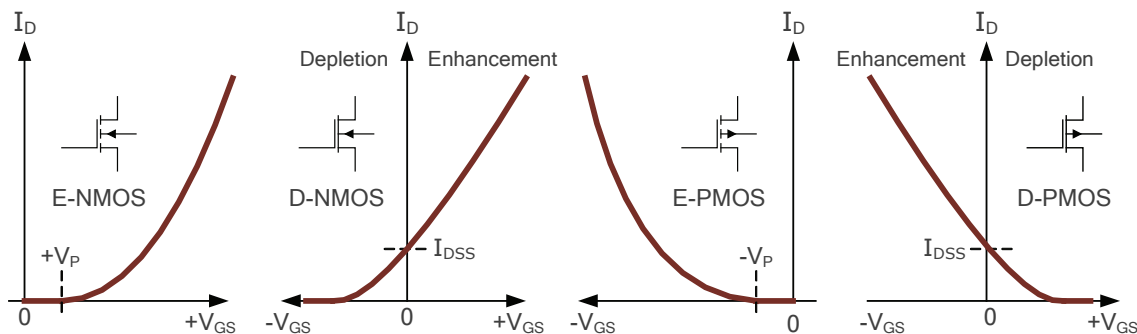
FIGURE 12. N-CHANNEL MOSFET CONSTRUCTION



Since the gate terminal is electrically insulated from the main current carrying channel, the MOSFET therefore has an extremely high input impedance. This means then that the MOSFETs gate current I_G is essentially zero for DC biasing configurations, hence the name insulated-gate FET (IGFET). V_{DS} and V_{GS} provide the necessary biasing voltages.

As with the JFET, transconductance curves can be plotted to show the characteristics of drain current, I_D for variations in gate-source voltage, V_{GS} for a fixed value of V_{DS} in the saturation region for all MOSFET types.

FIGURE 13. MOSFET CHARACTERISTICS CURVES



For the N-channel enhancement-mode MOSFET, (E-NMOS) I_D will be zero (channel-closed) when V_{GS} is less than V_p and increases with positive values of gate voltages.

The N-channel depletion-mode MOSFET, (D-NMOS) operates in its depletion-mode when V_{GS} is negative and closes the channel completely ($I_D = 0$) if V_{GS} is made sufficiently negative in value.

Depletion MOSFET's are generally used as amplifiers while Enhancement MOSFET's are generally used as switches

When V_{GS} is positive in value, the MOSFET changes and operates in enhancement-mode allowing the drain current to increase beyond I_{DSS} . Note that I_{DSS} is the drain current value that flows when $V_{GS} = 0$ and not the maximum drain current possible.

For the P-channel enhancement-mode MOSFET, (E-PMOS) I_D will be zero (channel-closed) when V_{GS} is more positive than V_p and increases with negative values of gate voltages.

The P-channel depletion-mode MOSFET, (D-PMOS) operates in the enhancement-mode when V_{GS} is negative allowing the drain current to increase beyond I_{DSS} and in the depletion-mode when V_{GS} is positive. For the D-PMOS FET, the conductive channel closes completely ($I_D = 0$) if V_{GS} is made sufficiently positive in value.

7. MOSFET OUTPUT CHARACTERISTICS CURVES

As with the JFET device, we can show the behaviour of the channel current by creating a set or family of output characteristics curves showing the direct relationship between the channel current I_D and different values of gate-source voltage V_{GS} . These I-V graphs also show the way the drain current behaves for different values of drain-source voltage, V_{DS} .

When V_{DS} is very small, little current flows through the MOSFET behaving like a voltage-controlled resistor in its linear or triode region.

The I-V characteristics of the N-channel Enhancement MOSFET of Figure 14. shows that the drain current is enhanced by a positive gate-to-source voltage. For the P-channel E-MOSFET the drain current is enhanced by a negative gate voltage. Thus the channel conductance is a function of the amplitude of the input signal.

FIGURE 14. ENHANCEMENT MOSFET OUTPUT CHARACTERISTICS CURVES

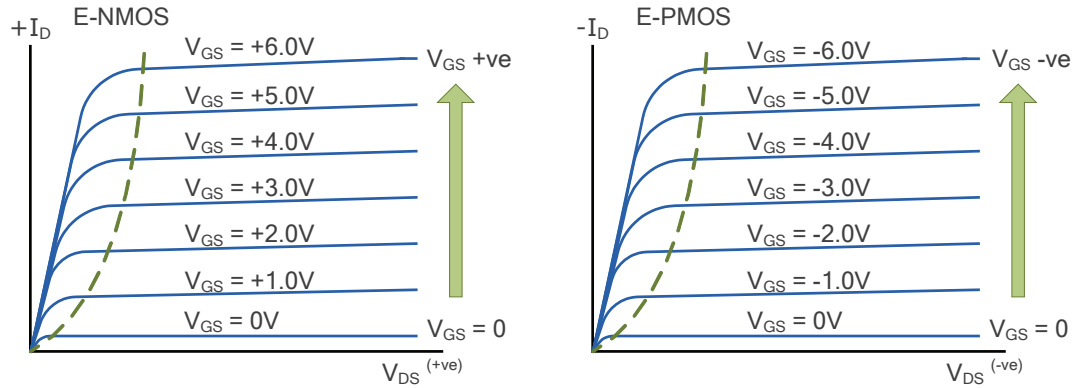
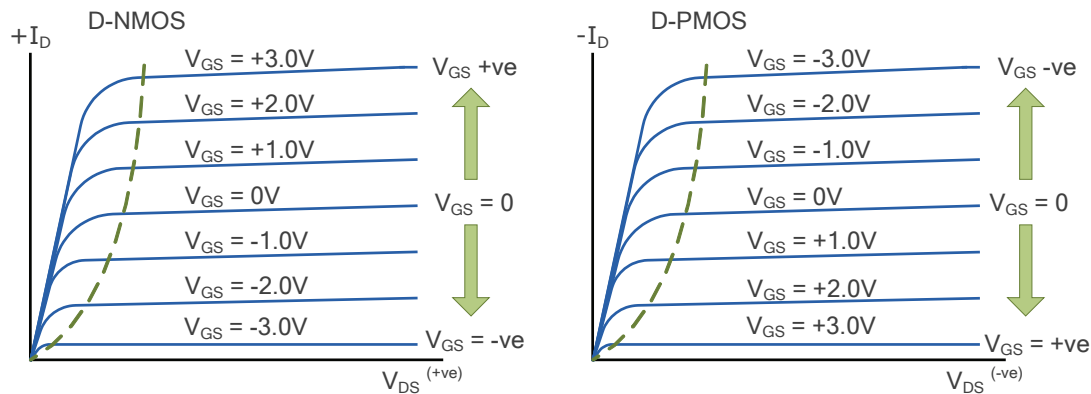


FIGURE 15. DEPLETION MOSFET OUTPUT CHARACTERISTICS CURVES



The I-V characteristics of the N-channel Depletion MOSFET of Figure 15, show that the drain current is enhanced by a positive gate voltage and reduced by a negative gate voltage. For the P-channel D-MOSFET the drain current is enhanced by a negative gate voltage but reduced by a positive gate voltage. Thus the channel conductance is a function of the polarity and the amplitude of the input signal.

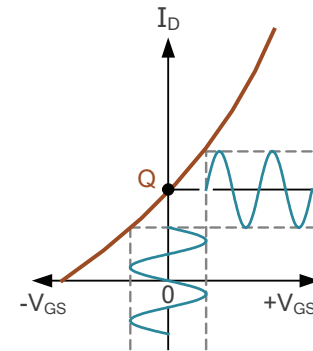
Then we can summarised the effect of different values of gate-source voltage, V_{GS} on the conductive channel in the following switching table.

TABLE 2. MOSFET GATE SWITCHING TABLE

MOSFET TYPE	$V_{GS} = +ve$	$V_{GS} = 0$	$V_{GS} = -ve$
N-channel Depletion Mode	ON	ON	OFF
N-channel Enhancement Mode	ON	OFF	OFF
P-channel Depletion Mode	OFF	ON	ON
P-channel Enhancement Mode	OFF	OFF	ON

You may have noticed that the depletion-mode MOSFET can operate with either a positive or a negative gate-source voltage and because of this it is commonly used in amplifier circuits with the Q-point set to correspond when $V_{GS} = 0$.

FIGURE 16. DEPLETION MODE Q-POINT CURVE



Thus, if we restrict V_{GS} to small variations around the I_{DSS} Q-point, the relationship between variations in V_{GS} and the resultant variations in I_D will be linear. So when the amplifiers input signal goes positive, I_D increases above I_{DSS} and when the input signal goes negative, I_D decreases below I_{DSS} .

Enhancement-type MOSFETs however, cannot be biased using the depletion-modes zero-biasing technique because the gate-source voltage V_{GS} must be greater than V_p to open the channel allowing drain current to flow.

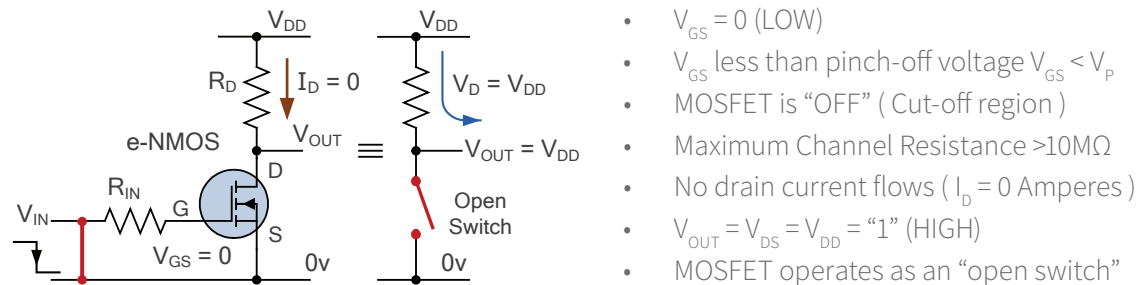
This then makes the enhancement-mode MOSFET extremely useful as an electronic switching device since the input gate voltage can be either low or high and well above the pinch-off voltage, V_p . This allows the MOSFET to operate between its cut-off and ohmic regions for ON-OFF switching applications.

8. THE MOSFET As A SWITCH

N-channel enhancement MOSFET's (e-NMOS) are the most widely used FET device for switching applications as their biasing arrangement can be very simple to achieve. For example, the drain terminal is always biased positive relative to the source terminal. While the input gate-to-source voltage V_{GS} which is used to control the conductivity of the channel is also a positive signal thus turning the device "ON" or "OFF".

Then for the N-channel enhancement MOSFET to operate as a solid state switching device, it is operated between its cut-off region, when zero current flows in the channel and its saturation region, when the transistor is fully conducting current and the drain-source voltage is large.

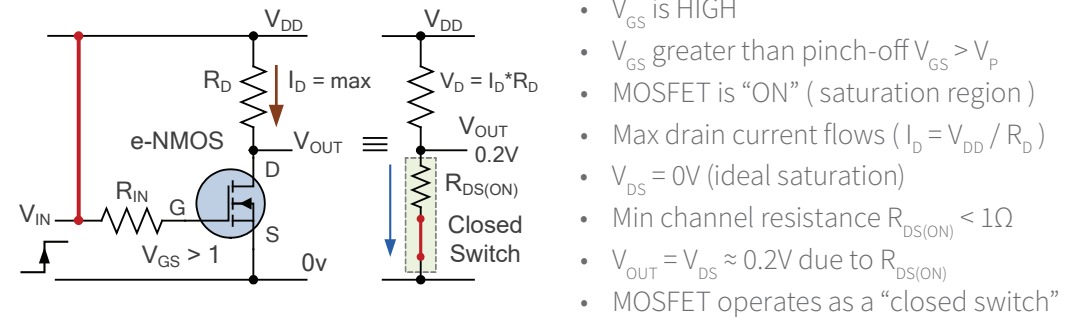
FIGURE 17. OPEN CIRCUIT (SWITCH) CONDITION (CUT-OFF)



When $V_{GS} = 0$ volts, the channel is closed between drain and source as $V_{GS} < V_p$ and so the device is switched "OFF" so $I_D = 0$. When using the e-NMOS device as a switch, the channel resistance in this OFF state is extremely high, typically over $10M\Omega$, which for most applications is effectively an open circuit condition.

The e-NMOS FET will remain indefinitely in this OFF condition as long as V_{GS} is zero or negative in value. For the complementary P-channel enhancement MOSFET, the gate potential must be made more positive with respect to the source.

FIGURE 18. CLOSED CIRCUIT (SWITCH) CONDITION (SATURATION)



When V_{GS} is made positive (gate positive relative to source), and greater than the pinch-off voltage V_p , (typically 1-to-2 volts for an e-NMOS FET) the conductive channel opens and current flows between the drain and source terminals. Generally a V_{GS} value much larger than V_p is used to turn the device "ON" completely.

Even though fully-ON, the channel resistance does not drop completely to zero (0Ω) due to the ON resistance $R_{DS(ON)}$ of the conducting channel resulting in a voltage drop of typically less than 1 volt across the drain-source terminals, V_{DS} .

Thus a lower $R_{DS(ON)}$ value for the channel resistance is a desirable parameter for a MOSFET as it helps to reduce the channels effective saturation voltage ($V_{DS(sat)} = I_D * R_{DS(ON)}$) and the device will therefore operate at a cooler temperature.

Power MOSFETs generally have a $R_{DS(ON)}$ value of less than 0.01Ω . This allows power FET's to run cooler, extending their operational life span.

$R_{DS(ON)}$ is the resistive value of the semiconductor channel when conducting current

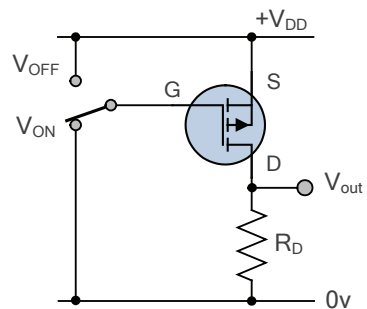
Thus by applying a suitable drive voltage to the gate of an FET, the resistance of the drain-source channel, $R_{DS(ON)}$ can be varied from an "OFF-resistance" of many hundreds of $k\Omega$, effectively an open circuit, to an "ON-resistance" of less than 1Ω , effectively acting as a short circuit.

9. THE P-CHANNEL MOSFET SWITCH

As well as the N-channel MOSFET used as a switch where the MOSFET is placed between the load and the ground allowing the MOSFET's gate drive or switching signal to be referenced to ground (low-side switching).

For P-channel enhancement-mode MOSFET the load is connected directly between the drain terminal and ground as shown in Figure 19.

FIGURE 19. P-CHANNEL MOSFET CONFIGURATION



The P-channel MOSFET operates in exactly the same way as the previous N-channel device except that the voltage polarities are reversed.

For the E-PMOS FET conventional flow of drain current is in the negative direction so a negative gate-source voltage greater than $-V_p$ is applied to switch the FET transistor "ON".

Thus the E-PMOS device is connected between the load and the positive supply rail (high-side switching) while the drain terminal is connected to the load. That is to drive the E-PMOS FET properly, the gate voltage must be referenced to its source.

Because the P-channel MOSFET is effectively "upside down" with its source terminal tied to the positive supply $+V_{DD}$, when the switch goes LOW, the MOSFET turns "ON" and when the switch goes HIGH the MOSFET turns "OFF".

Thus to turn an E-PMOS FET fully "ON", an input voltage lower than the source voltage and greater than $-V_p$ must be applied to the gate terminal, meaning the voltage at the gate, relative to the source, must be negative. In other words, to turn ON an N-channel MOSFET requires a positive gate-source voltage, whereas the P-channel MOSFET requires a negative gate-source potential.

P-channel and N-channel are complementary and operate with inverted voltage polarities

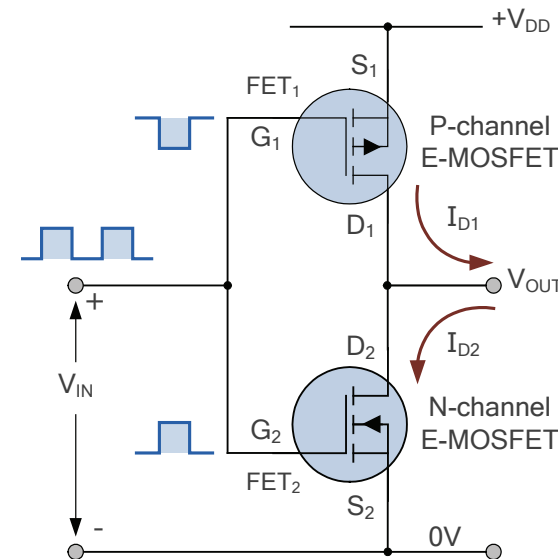
One advantage of this upside down connection for a P-channel enhancement mode MOSFET switch is that it allows us to connect it in series with an N-channel enhancement mode MOSFET to produce what is called a complementary MOS or CMOS switching arrangement.

10. COMPLEMENTARY MOS SWITCHING

We can construct a complementary MOS (CMOS) switching device by connecting a PMOS and an NMOS device together in series so that the P-channel device has its source connected to V_{DD} (a positive voltage), and the N-channel device has its source connected to ground (a negative voltage) as shown in Figure 20.

The gate terminals of the two MOSFETs are connected together to produce a common input. The drain terminals of the two MOSFETs are joined together to produce a common output.

FIGURE 20. CMOS CONFIGURATION



When the input is LOW the P-channel MOSFET is switched-ON as its gate-source junction is negatively biased. The output goes HIGH as it is connected to the positive $+V_{DD}$ supply rail (high-side switching). Thus drain current, I_{D1} flows out (current-source) from the MOSFET combination.

When the input is HIGH, the P-channel device switches-OFF and the N-channel device switches-ON as its gate-source junction is positively biased. The output goes LOW as it is connected to ground and 0V, (low-side switching). Thus drain current, I_{D2} flows into (current-sink) the MOSFET combination.

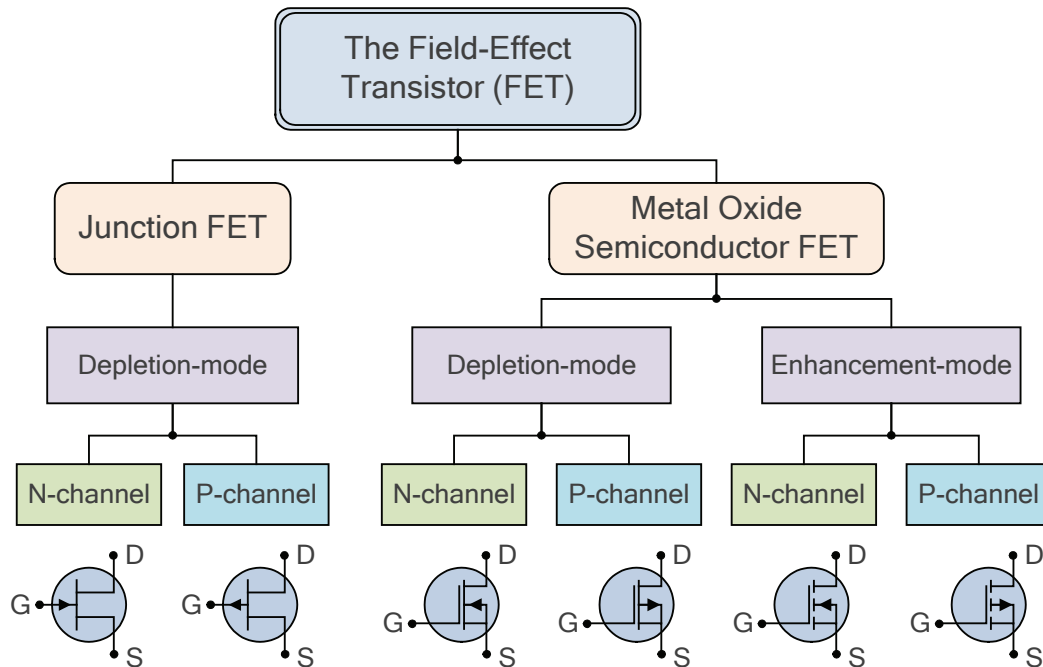
Then the P-channel MOSFET is used to switch the positive supply to the output, while the N-channel MOSFET is used to switch the negative supply to the output. Thus, either the PMOS or the NMOS device is ON while the other is OFF

This current-sink, current-source switching action makes the CMOS switch very useful for controlling the direction of DC motors connected to the output terminal. Therefore because of the switching action of the two FET's, the output is LOW when the input is HIGH, and HIGH when the input is LOW effectively inverting the input signal.

11. FIELD EFFECT TRANSISTOR COMPARISON

We can summarise what we have learnt in this ebook by listing the different types of *Field Effect Transistor* and their operating modes in the following chart.

FIGURE 21. FIELD EFFECT TRANSISTOR TYPES



The required voltage biasing of the Gate for both the junction field effect transistor, (JFET) and the metal oxide semiconductor field effect transistor, (MOSFET) configurations are given in the following table.

TABLE 3. FET BIASING VOLTAGES

Type	Junction FET (JFET)		Metal Oxide Semiconductor FET (MOSFET)			
	Depletion Mode		Depletion Mode		Enhancement Mode	
Bias	ON	OFF	ON	OFF	ON	OFF
N-channel	0V	-ve	0V	-ve	+ve	0V
P-channel	0V	+ve	0V	+ve	-ve	0V

[End of this Field Effect Transistor eBook](#)

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With the completion of this Field Effect Transistor eBook you should have gained a basic understanding and knowledge of FET's. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering. In ebook 15 you will learn about thyristors and triac's which are power electronic devices.

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